ABHI SHAH

Austin, TX | abhishekshah7@gmail.com | 737-203-4804 | www.linkedin.com/in/ashah-mecheng | Portfolio

COMPETENCIES

Tools And Software: SolidWorks, Creo, Onshape, CATIA, PDM, Windchill, C++, HTML, JavaScript, Figma, Raspberry Pi, esp32, Arduino

Skills: Sheet Metal Fabrication, Machining, Die Casting, Aluminum Extrusions, 3D Printing, Surfacing, Top-Down Modeling, DFM, DFA, FEA, Tolerance Stack-Up Analysis, Injection Molding, FMEA, Validation Testing

Standards: NEMA 4X, IPXX Compliance, NADCA Standards, IS Standards, CSA Standards, MIL-STD

PROFESSIONAL EXPERIENCE

Whisker (In-home Robotics and Automation for Pets) — Austin, Texas

System Engineer — Jul 2026 – Present

- Own the technical success of Whisker's next-generation connected robotics platform end-to-end, driving alignment and execution across Mechanical, Electrical, Firmware, and Software Engineering
- Serve as the central technical decision-maker keeping architecture, requirements, and trade-offs coherent as the product moves from concept to production
- Partner with cross-functional engineering leads to de-risk integration points early, catching interface issues before they become costly downstream
- Drive design reviews and prioritization decisions, and act as the escalation point for cross-discipline technical blockers to keep the program on schedule without compromising quality

Lead Mechanical Engineer — May 2025 – Jun 2026

- Evaluated the competitive landscape and developed new architecture to better align the product with company goals. Reduced COGs by 50% while improving user experience
- Lead engineering of 2 products in parallel across mechanical, electrical, firmware and manufacturing
- Mentored several junior engineers through delegating tasks, providing support and developing processes

Pipedream Labs (Robotics and Automation Startup) — Austin, Texas

Product Owner / Sr. Mechanical Engineer — Aug 2023 – May 2025

- Owned end-to-end development of a core subsystem for a multi-million dollar underground automation project installed at a $500M restaurant chain in Dallas, TX. Planned and executed from architecture to deployment.
- Mentored junior engineers by delegating tasks and providing guidance and guardrails
- Implemented comprehensive SOPs for CAD document management, introducing engineering best practices
- Identified and resolved major inefficiencies in Onshape sheet metal design workflows and improved performance by 90% on comparable files for the entire team
- Created an internal tool to automate revision-controlled BOM creation and management, adding preventative controls prior to production

Pump Studios (Engineering Consulting) — Austin, Texas

Product Owner / Sr. Mechanical Engineer — Mar 2022 – Aug 2023

- Owned proof of concept to PVT for multi-million dollar project currently available in the market
- Led cross-functional collaboration while successfully negotiating trade-offs between features, timeline and budget
- Developed products for heavy industrial automation, consumer electronics and manufacturing fixtures

Catalyst Design Consulting Inc. (Engineering Consulting) — Calgary, Alberta

Founder / Sr. Mechanical Design Engineer — May 2018 – Feb 2022

- Founded an engineering consulting startup, averaging 6 figures in revenue per year
- Designed medical devices with the University of Miami. Co-inventor of U.S. Patent # 11957569 , 20200276010
- Developed a IP67 rated hand held gas detector using two shot Injection molding with slider mechanism for 10,000 units/year production and tested to Intrinsic Safety (IS) and CSA standards
- Reduced consumer electronic product cost by 40% with product redesign and overseas vendor management

Nureva Inc. (Audio/Video Conferencing Startup) — Calgary, Alberta

Sr. Mechanical Engineer — Mar 2021 – Sep 2021

- Achieved 30% cost reduction on flagship product by optimizing part count, material selection and manufacturing
- Directed the migration of manufacturing operations from China to Malaysia, enhancing supply chain
- Implemented and managed PDM system that streamlined collaboration and improved workflow efficiency

SEE PROJECT DETAILS IN PORTFOLIO

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Novatel Inc. (GPS and GNSS Systems) Calgary, Alberta
Mechanical Engineer & Manufacturing Engineer Apr 2015 – May 2018

- Conducted rigorous reliability testing on PCBAs and electromechanical packaging including HALT, drop, ingress, humidity, vibration, and bump tests—to validate product durability and ensure compliance per MIL-STD
- Reviewed PCBA designs for manufacturability and Developed PCB/PCBA manufacturing processes to get reliability of 99.96%

EDUCATION

Bachelor of Engineering (B.Eng.), Mechatronics Oshawa, Ontario
Ontario Tech University 2010 – 2014

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